Viewing Multiple Interactive Plots with {Plotly} and {Trelliscopejs}

A Lipidomic Data Example

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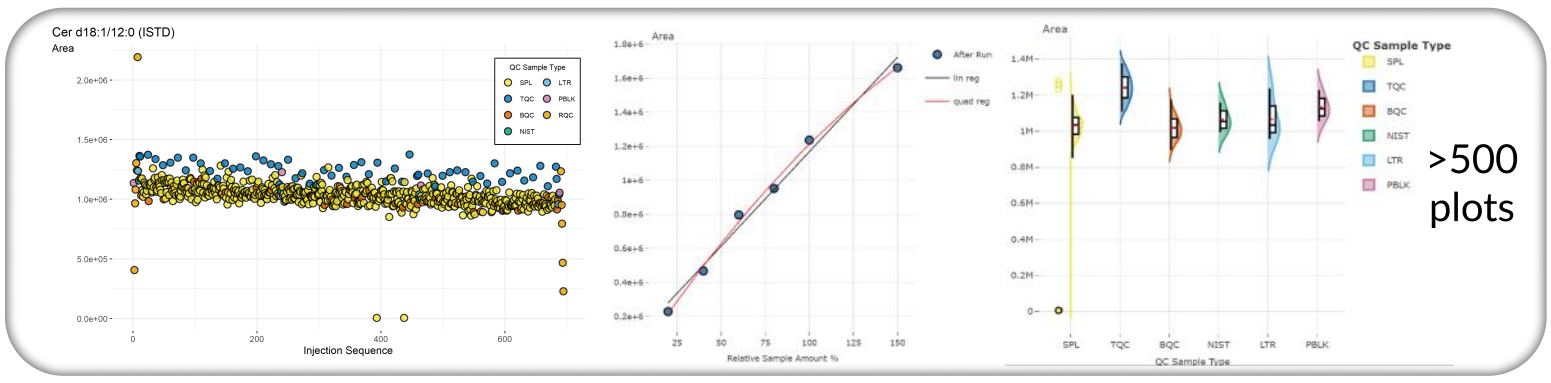
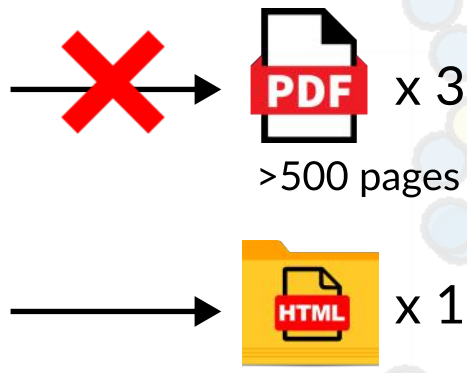
Poster design inspiration from Cara Thompson (Twitter: @carathompson)

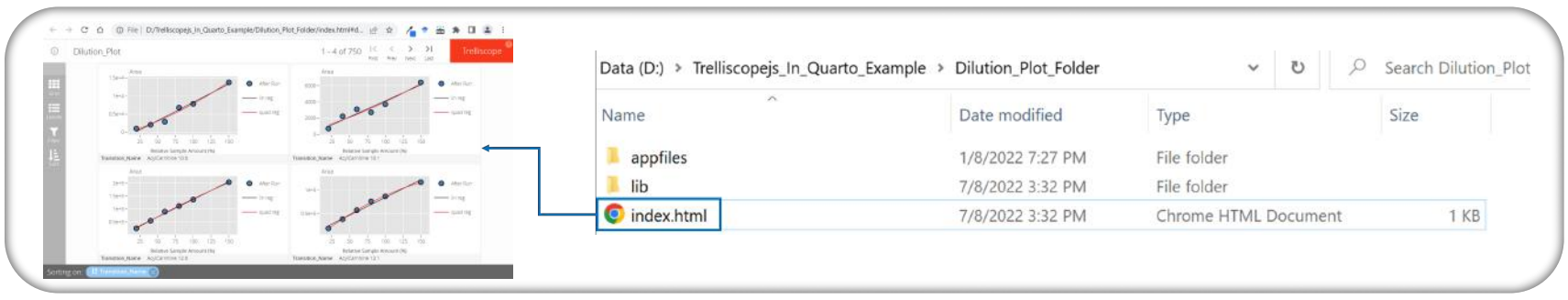
#1 Motivation

In the mass spectrometer, variation of signal between samples can come from biological or external sources, such as contaminant ions. Quality control (QC) samples are used to track for these variations associated with sample preparation and data acquisition. For a given measurement, an injection sequence chart is used to view the variation. These plots are converted as pages in a pdf file.

However, today's targeted lipidomics workflow using liquid chromatography can measure up to over five hundred molecules in a few thousand samples. Moreover, new QC sample types also give rise to new plots. Having a data analyst to look at many pdf files full of static plots has its limitations. **An output of interactive plots that can be distributed easily is required**.

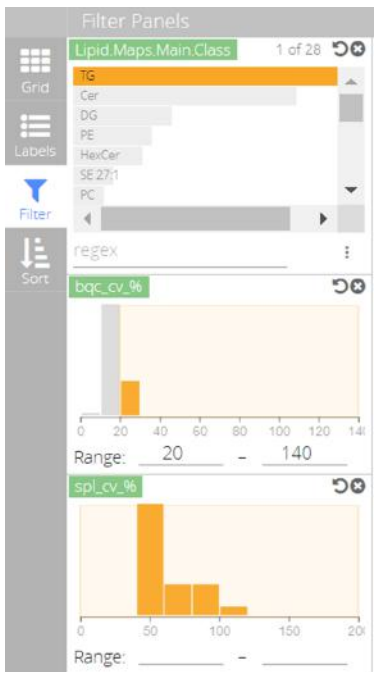


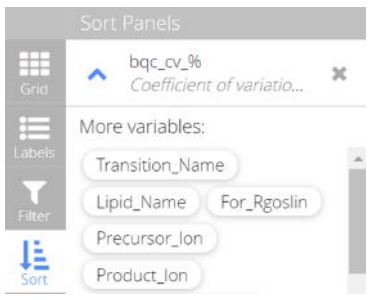


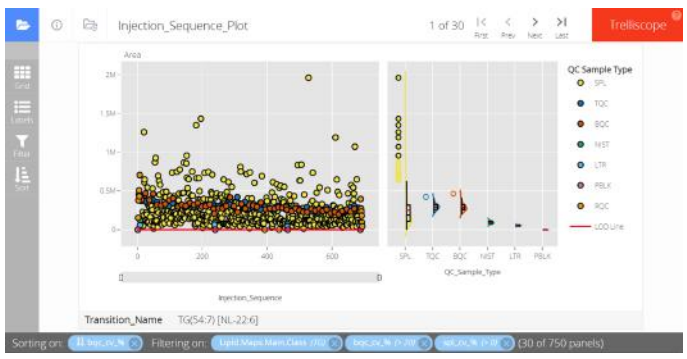


#2 Advantages

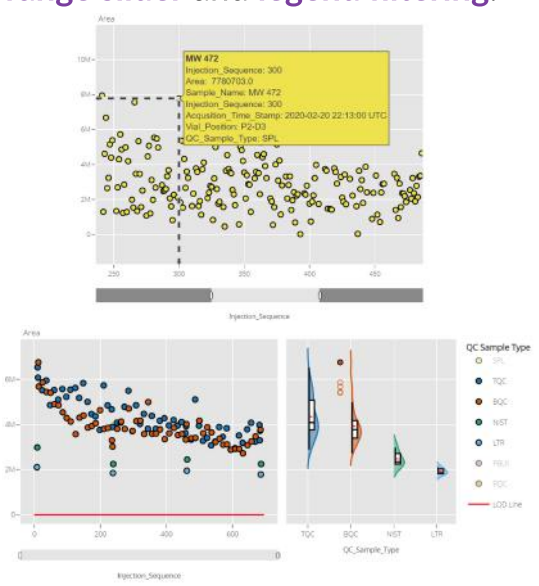
Outputs from {Plotly} and {Trelliscopejs} are able to give displays of interactive plots that can be **grouped**, **filtered** and **sorted**.



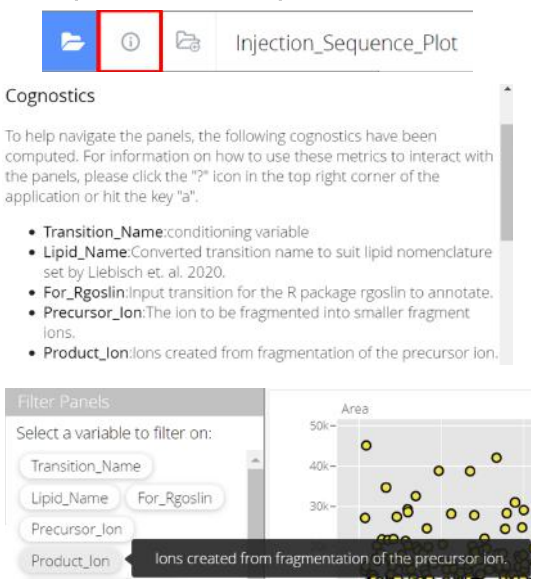




Data points can be analysed using **range slider** and **legend filtering**.

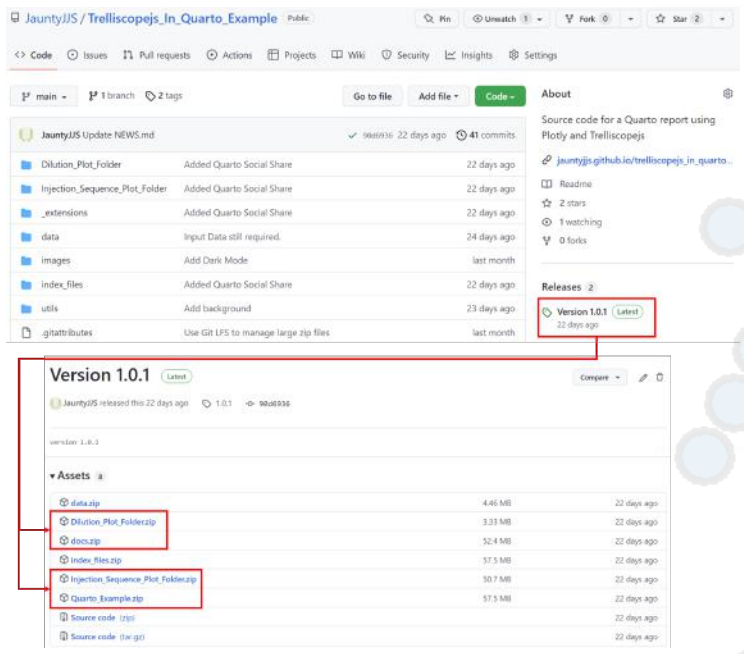
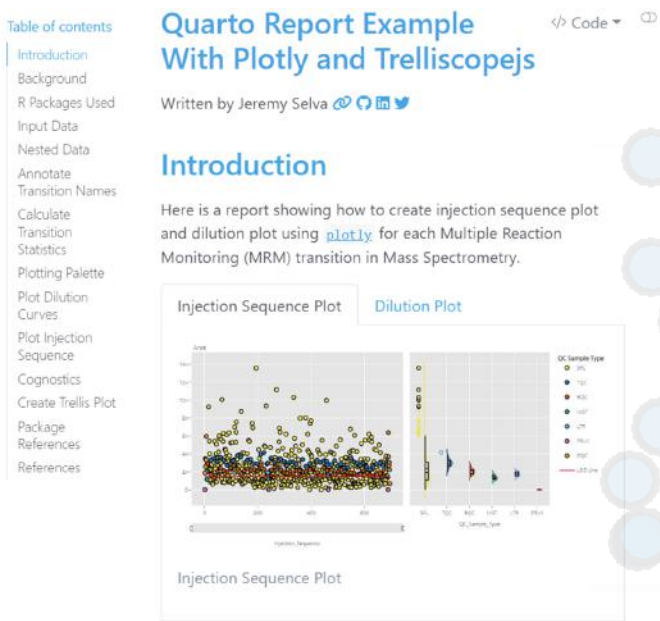


Meta information can be provided to improve user experience.



#3 Resources

Examples in poster can be found on GitHub.



Walkthrough guide can be found on this webpage.